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Systematic Review of Primary Healthcare Interventions to Improve Diabetes Outcomes in Minority Ethnic Groups

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Abstract: Objectives: To systematically review the effectiveness of primary care interventions on glycaemic control and cardiovascular risk factors in minority ethnic groups with diabetes. **Research design and methods:** We searched electronic databases, the Cochrane Library, and research registers to December 2006, using multiple search terms related to ethnicity and diabetes. We examined bibliographies of retrieved articles and corresponded with authors. We included randomized controlled trials, controlled clinical trials, and cohort studies. Two reviewers independently assessed study eligibility and quality. **Results:** Nine studies including 2565 patients met our inclusion criteria. Two main models of care were identified: (1) case management, with specialist diabetes nurses and community health workers and (2) the use of the services of link workers from minority ethnic groups to guide people with diabetes. Heterogeneity of the studies prevented us from carrying out a meta-analysis. Case management improved glycaemic control (reduction in HbA1c range −0.5% to −1.75%). Small but statistically significant reductions in other cardiovascular risk factors were reported with both models. **Conclusions:** In minority ethnic groups with diabetes, case management improves glycaemic control and cardiovascular risk factors and link workers improve cardiovascular risk factor control. However, their relative effectiveness, cost, and sustainability of changes over time warrant further evaluation. **Key words:** *case management*, *diabetes*, *health service interventions*, *link worker*, *minority ethnic*, *primary care*

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RESEARCH has extensively documented the pervasiveness of racial and ethnic disparities in healthcare in many developed countries, including the United States (Smedley et al., 2002). and the United Kingdom (Arai & Harding, 2002), despite calls from their governments to develop policies to eliminate inequities in health and provision of healthcare (Acheson et al., 1998; Smedley et al., 2002). Minority ethnic groups, many of whom are already at higher risk of developing chronic conditions such as diabetes, cardiovascular disease, and asthma than the general population (Harding, 2003; Meyer et al., 1994; Netuveli et al., 2005), often have poorer access to services for such conditions in both primary and secondary healthcare settings (Aspinall & Jacobson, 2005; Feder et al., 2005). The growing burden of illness from

diabetes is among the most pressing healthcare problems faced by these nations (King et al., 1998) and represents an area where disparities in prevalence and healthcare between ethnic groups are greatest (Kramer et al., 2004) and are predicted to grow to epidemic proportions among some minority ethnic groups (Mainous et al., 2006).

A number of educational strategies targeted at healthcare professionals and innovative methods of organizing and providing healthcare have been proposed to improve the recognition and management of diabetes in minority ethnic groups in primary care. However, the clinical effectiveness and cost-effectiveness of these strategies are unclear (Beach et al., 2004; Eccles et al., 2003). Quality improvement strategies in the management of diabetes mellitus have been heralded as a means of improving access to services and better management for all people with diabetes but improvements in glycaemic control and other risk factors may require a focus on minorities (Sequist et al., 2006; Trivedi et al., 2006).

We conducted a systematic review to examine the effectiveness of health service interventions in primary care on clinical outcomes for diabetes in minority ethnic groups. Our research question was "do current health service interventions improve glycaemic control and reduce cardiovascular risks in diabetic patients from minority ethnic groups?"

METHODS

Data sources

We performed a comprehensive search of several medical and psychological databases including MEDLINE, EMBASE and CINAHL, and the Cochrane Database of Systematic Reviews, the Cochrane Library, research registers and grey literature, from inception to December 2006, using MeSH and key words: minority group or ethnic group or ethnic and racial groups or ethnic group or minority group or immigrants or Indian or Pakistani or Bangladeshi or Asian continental ancestry group or African continental ancestry group, and diabetes mellitus and relevant subject subheadings. We examined the reference lists of retrieved articles and contacted experts in the field.

Study selection

Two reviewers independently screened the abstracts of all citations and retrieved studies that met the inclusion criteria (Fig 1). Given the complex nature of educational and organizational interventions, it is not appropriate to consider only evaluations by randomized controlled trial (Eccles et al., 2003). We included randomized controlled trials, controlled clinical trials, controlled before-and-after studies, and interrupted time-series studies reporting the effect of interventions on diabetes control and related health outcomes in minority ethnic groups. Because the review was focused on examining interventional models of healthcare delivery potentially relevant to primary care for minority groups in the United States and the United Kingdom, we excluded articles not in English and articles focused on specialist pediatric or adolescent services. If we found insufficient information in the abstract to decide on its eligibility, the full article was retrieved for review. Where necessary, authors of studies were contacted to clarify details or obtain additional information. Two reviewers independently abstracted data on participant characteristics, study design, interventions used, outcomes, and level of evidence ("Levels of Evidence," 2001). Discrepancies were resolved by arbitration by a third member of the team. Studies that met inclusion criteria were graded for methodological quality using an adapted quality score reported by Beach et al. ranging from 0 to 40 where higher scores indicate better quality (Beach et al., 2004; Undertaking Systematic Reviews of Research on Effectiveness, 2001).

Outcomes

Primary outcome measure was improvement in glycaemic control (reduction in HbA1c). Additional outcome measures included reduction in cardiovascular risk factors (systolic and diastolic blood pressure, total cholesterol, and triglyceride levels).

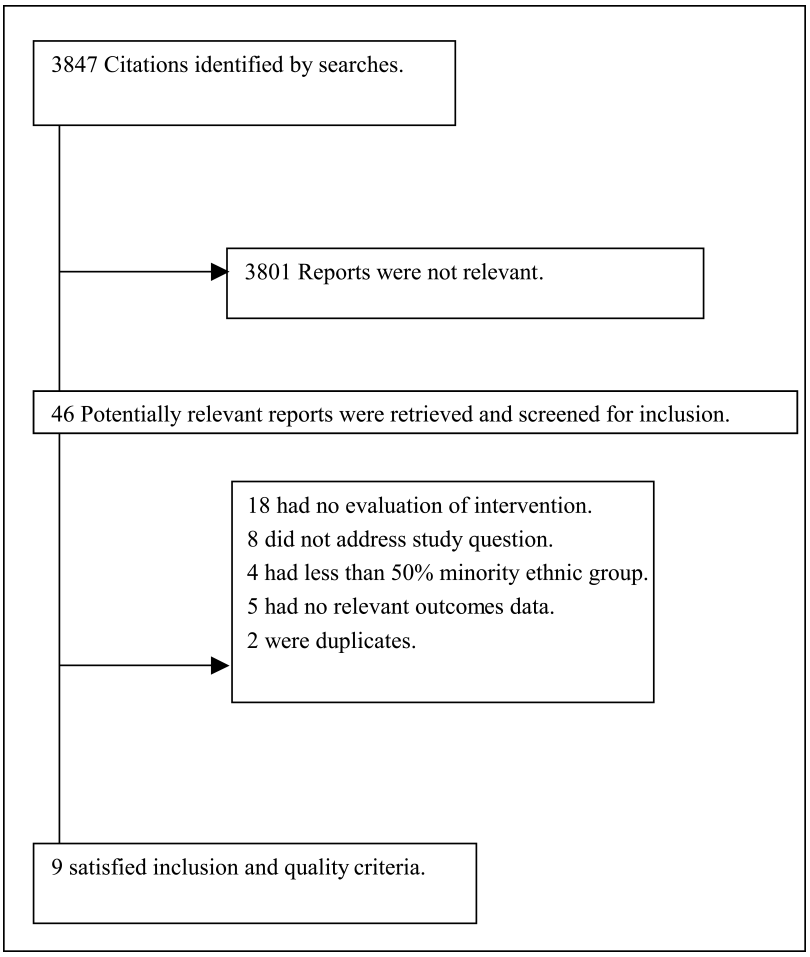

Figure 1. QUORoM statement flow diagram of included and excluded studies. Adapted from http://www.consort-statement.org/QUOROM.pdf.

Statistical analysis

Where suitable data were available, we tested for the suitability of quantitative synthesis. The heterogeneity between studies was statistically significant ($P < .001$) and was not amenable to statistical correction. Therefore, we present the results qualitatively below.

RESULTS

Our search yielded 3847 studies, of which 96 were retrieved for further review after screening titles and abstracts; 50 of these were found unsuitable on reading the full paper and the remaining 46 went on to have detailed assessment (Fig 1) (Moher et al., 1999). Nine articles met our criteria for quality and inclusion and included 2565 participants in total. The number of participants in each included study ranged from 105 to 713 (Tables 1 and 2).

Study characteristics

Four of the studies were from the United States, 4 from the United Kingdom, and 1 was from the Netherlands. The studies from the United States covered either African American (1) or African American and Hispanic (3) ethnic groups. Two of the studies from the United Kingdom covered British Indians, and 2 studies covered British Pakistanis. The study

Table 1. Characteristics of studies grouped by nature of the intervention*

Source	Country	Ethnic groups	No. of subjects	Mean Age, y	Study design	Key interventions	Time frame, mo	Quality score (Beach et al., 2004)	Outcomes
Case management									
California Medical Type 2 diabetes study group2004 (The California Medi-Cal Type, 2004)	US	Hispanic African American	358	57	RCT	Extensive case management intervention—nurse, dietician & endocrinologist input	25	28	Hba1c
Davidson (2003)	US	African-American– 2% Hispanic– 92%	713	52	CCT	DMCP—Diabetes Managed Care Programme, delivered by nurses and pharmacists using standard algorithms and protocols	6	30.5	HbA1c LDL cholesterol,† triglycerides ACE inhibitor or ARB treatment of microalbuminuria or proteinuria
Gary, 2003 (Johns Hopkins – Project Sugar)	US	African-American	149	58	RCT with 4 parallel arms	Patient counseling regarding self-care practices and physician reminders, through nurse case managers and/or healthcare workers	24	29	HbA1c and SBP, lipids
Middelkoop (2001)	Netherlands	Surinamese South Asian	113	52	RCT followed by a CBA	Intensive guidance and patient education from diabetes nurse specialist and dietician	6	No score	HbA1c, lipids, BMI
Vyas (2003)	UK	South Asian	118	Nk	RCT	4 or more rotating visits per year by diabetes nurse specialist, dietician, or chiropodist working with practice nurses in each surgery	12	28	Knowledge awareness and self-management scores for diabetes

(*continues*)

Table 1. Characteristics of studies grouped by nature of the intervention* *(Continued)*

Source	Country	Ethnic groups	No. of subjects	Mean Age, y	Study design	Key interventions	Time frame, mo	Quality score (Beach et al., 2004)	Outcomes
Link worker									
Hawthorne (1997)	UK	Pakistani	201	53	RCT	Link worker providing structured pictorial teaching program based on the use of culturally appropriate pictorial flashcards and individual tuition	6	33	HbA1c Cholesterol Knowledge scores, changes in self-caring behaviors and attitudes to diabetes
Hawthorne (2001)	UK	Pakistani women	105	55	RCT	Individual structured diabetes health education by a link worker, with pictorial flash cards	6	33	Hba1c knowledge assessment scores
O'Hare (2004)	UK	South Asian	361	59	RCT	2 Asian link workers, additional practice nurse session per week, and community diabetes specialist nurse	12	35	BP HbA1c Total cholesterol
Reminder cards									
Echeverry (2003)	US	Hispanic African American	447	54	CCT	RC Reminder cards to prompts providers	12	26	Attendance for diabetic monitoring

*CCT indicates controlled clinical trial; RCT, randomised control trial; CBA, controlled before and after study.
Studies grouped by key intervention models: CM indicates case management; LW, link worker; RC, reminder cards; BP, blood pressure; SBP, systolic blood pressure; ACE, acetyl cholinesterase inhibitor; and ARB, angiotensin receptor blocker.

Table 2. Effect of health interventions on clinical outcomes for diabetes*

Source	Mean change in clinical outcomes between intervention and control groups over study period (95% confidence interval where given)†					Effect on other outcomes and comment
	Weighted mean differences in glycaemic control-HbA1c‡ (%)	SBP, mm Hg	DBP, mm Hg	Lipids, mmol/L	BMI	
Case management						
California Medical Type 2 diabetes study group (2004)	−0.87 (−0.91, −0.83)	0	−3†	TC −0.2†,§ LDL −0.4†,§ HDL −0.1†,§	0	Significant improvements in HbA1c, HbA1c goals, medication usage between intervention and control groups. Secondary outcomes not different between groups, but differed from baseline in intervention group.
Davidson (2003)	−1.00 (−1.75, −0.25)	–	–	–	–	Nurse case management effective in implementing ADA guidelines led to significant improvements in treatment of risk factors including LDL cholesterol† triglycerides† in this minority ethnic population of low socioeconomic status
Gary (2003) (Johns Hopkins – Project Sugar)	0.10 (−0.81, 1.01)	0	−5.6†	TGs = 0§	0	Combined NCM/CHW interventions may work to improve outcomes in "hard to reach" urban African Americans. Underpowered
Middelkoop (2001)	−0.5†	–	–	TC −0.56 mmol/L† TGs −0.34 mmol/L†	0	Significant reduction in HbA1c in RCT group. And reduction in HbA1c, cholesterol, but not BMI in CBA group. Development of culture-specific diabetes care can have a beneficial effect on metabolic control.
Vyas (2003)	–	–	–	–	–	Diabetic nurse specialists and allied professional visiting patients had no impact on scores for diabetes knowledge (0.14, −0.20 to 0.49) or awareness of diabetes (-0.05, −0.48 to 0.39). No data collected on biochemical parameters.

(*continues*)

Table 2. Effect of health interventions on clinical outcomes for diabetes* *(Continued)*

	Mean change in clinical outcomes between intervention and control groups over study period (95% confidence interval where given)†					
Source	**Weighted mean differences in glycaemic control-HbA1c‡ (%)**	**SBP, mm Hg**	**DBP, mm Hg**	**Lipids, mmol/L**	**BMI**	**Effect on other outcomes and comment**
Link worker						
Hawthorne (1997)	−0.34 (−0.95, 0.27)	–	–	0	–	Glycaemic control did not improve significantly but significantly improved knowledge† and self-caring behaviors† after 6 months.
Hawthorne (2001)	−0.56 (−1.39, 0.27)	–	–	–	–	Important intervention for women who do not speak English resulting in significantly improved knowledge† and self-caring behaviors† after 6 months. However outcomes in illiterate women were unchanged.
O'Hare (2004)	0	−4.6*	−3.4†	TC- 0.4mmol/L		Improvements in blood pressure among non-English speakers but not illiterate women
Reminder cards						
Echeverry (2003)	–	–	–	–	–	Patient-led reminders for diabetes care resulted in significant improvements in 2 ADA process measures: foot care and urine testing.

*Outcomes: SBP indicates systolic blood pressure; DBP, diastolic blood pressure; BMI, body mass index. Lipids: TC indicates total cholesterol; TGs, triglycerides; LDL, low density lipids; HDL, high density lipids; and ADA, American Diabetes Association guidelines. Dash indicates that outcome is not reported.

†Improvements in outcomes reported by authors as significant $P < .05$, where confidence limits are not reported.

‡Weighted by the number of subjects.

§Converted from US units mg/dL into standardized international units (mmol/L) using online millimol converter (http://www.crestor.info/19119_0.aspx?mid=41&l1=6&l2=0&ch=Millimol+Converter).

from the Netherlands covered South Americans from Surinam.

Models of intervention

The studies presented 2 main models of healthcare interventions, which we labeled as (1) *Case management model* and (2) *Link worker model*.

Case management model

Five of the studies provided evidence for this model, which is delivered within the existing service infrastructure, with an extra member of staff, appropriate staff training, and the use of standardized treatment algorithms, usually under the supervision of a diabetologist (Davidson, 2003; Gary et al., 2003; Middelkoop et al., 2001; The California Medi-Cal Type, 2004; Vyas et al., 2003). Case management was successful in improving glycaemic control (statistically significant reductions in HbA1c%) in 3 of 4 studies and small reductions in other cardiovascular risk factors including blood pressure (2 of 4 studies) and total cholesterol and triglyceride levels (Table 2).

Of the 5 studies that used the case management model, 1 was from the United Kingdom, 3 were from the United States, and 1 was from the Netherlands. This model was mainly delivered by specialist nurses. Three studies had a team consisting of a specialist diabetes nurse and a dietician; one study had just one full-time nurse, and one had a team comprising a nurse care manager and community health worker. In the UK study, the case management intervention consisted of 4 or more rotating visits per year by one of the staff (a diabetes specialist nurse, podiatrist, or dietician, working with the practice nurse) (Vyas et al., 2003). In addition, there was either an Urdu- or Hindi-speaking healthcare professional or interpreter present at each session. The diabetes specialist nurse, podiatrist, and dietician provided healthcare, trained practice nurses and then supervised them in delivering clinic sessions. The study focused on the impact of the intervention on diabetes self-management and awareness only, and found that it was not effective. In the US study from Johns Hopkins University (Gary et al., 2003), the intervention consisted of at least 3 visits per year with each of the nurse case manager and community health worker, respectively or together. The role of the nurse case manager was to provide 45-minute clinic sessions and/or telephone contacts; the community health worker provided 45- to 60-minute home visits and/or telephone contact. The combined intervention, which led to a reduction in diastolic blood pressure and triglycerides among the intervention group, may improve outcomes in "hard to reach" urban African Americans.

Two studies provided managed care, using a nurse (Davidson, 2003) or a team of nurses and dieticians (The California Medi-Cal Type, 2004) to a population of mainly African Americans and Hispanics. This was done using protocols and guidelines under the supervision of a diabetologist. Davidson et al. demonstrated improvement in the process measures reviewed, which met American Diabetic Association guidelines (Davidson, 2003). The Medi-Cal group found a reduction in BP and total cholesterol levels in the intervention group (The California Medi-Cal Type, 2004).

The study from the Netherlands used a specialist diabetes nurse and a dietician, whose roles were to provide diabetes education in the form of intensive guidance (4–7 visits) for 3 months, followed by sessions at longer intervals (Middelkoop et al., 2001). They developed an audio cassette on diabetes in Surinam-Hindi and 2 booklets: one on nutrition and the other on carbohydrate exchange, based on South Asian cooking, to promote these messages. This culturally specific health education was found to improve HbA1c, total cholesterol, and triglyceride levels.

The staff who delivered the case management model did not have any cultural or linguistic affinity with the communities they were serving (except in the case management study from the United Kingdom, where a multilingual healthcare professional or interpreter was present at each clinic session). The level of training provided to the "case management workers" varied between the studies. Only 2 of the 5 studies specified what additional

training, if any, was given to the staff. In the Diabetes Managed Care Programme, special training was given to the full-time nurses employed for the study, to follow carefully drawn protocols and algorithms (Davidson, 2003). In the study from the Netherlands, the diabetes specialist nurse and dietician were trained to improve their knowledge of South Asian cultural and culinary traditions (Middelkoop et al., 2001).

Link worker model

Three studies provided evidence for this model that involves trained members from the minority ethnic communities who provide advice to improve patients' access to health services. In contrast to the healthcare professional staff used in case management approaches, link workers are lay people with no medical background and provide links to healthcare services, and information about diabetes management and care pathways (Hawthorne, 2001; Hawthorne & Tomlinson, 1997; O'Hare et al., 2004). Small improvements in glycaemic control (reduction in HBa1c) reported with link workers did not reach statistical significance and only one study found small, albeit significant, reduction in cardiovascular risk factors including blood pressure and serum lipids (O'Hare et al., 2004).

All of the 3 studies that used the "link worker" model were from the United Kingdom. There is no standard job description or role of the link workers seen in these studies. However, a careful comparison reveals some common characteristics. In the UK Asian Diabetes Study (UKADS), the link workers' role was to provide interpretation in the extra clinic session held by the diabetes specialist nurse every week in the intervention practices to enhance patient understanding and compliance. The link workers also contacted patients before and after clinics to encourage attendance and organized educational sessions for patients (O'Hare et al., 2004).

In the 2 studies from Manchester, the role of the link worker was to deliver a structured education package, using flashcards in one-to-one 20-minute teaching sessions. The sessions were conducted in a hospital clinic, general practitioner clinic, or the participant's home. The link worker did 4 to 6 sessions per week (Hawthorne, 2001; Hawthorne & Tomlinson, 1997). All link workers in the 3 studies from the United Kingdom spoke the main languages of the communities that they served. None of them had any medical or nursing education, and none of the studies specified their education levels. In the UKADS, the 2 link workers are only described as being multilingual (O'Hare et al., 2004). In the 2 studies from Manchester, the link worker was a married woman from the Pakistani community who had a background in teaching and religious studies (Hawthorne, 2001; Hawthorne & Tomlinson, 1997).

The level of training provided to the link workers varied between the studies. In the UKADs, the training provided to link workers is not specified, and their role limited to liaison and interpretation. In the Manchester studies, the link workers were given intensive in-house training at the Manchester Diabetes Centre to provide health education. This was the only study we found reporting the cost of the intervention (£ 365 per diabetic patient) (Hawthorne & Tomlinson, 1997)

Process intervention

One study evaluated the effectiveness of a low-literacy, diabetes reminder card in enhancing certain diabetes process care measures (Echeverry et al., 2003). Although low-literacy reminder cards did improve the ordering of process measures, they were not as effective as the utilization of progress forms with specific diabetes-related measures.

CONCLUSIONS

This systematic review has identified 2 main models of interventions aimed at improving clinical health outcomes in people with diabetes from minority ethnic groups. Of these, the case management model resulted in significant improvement in glycaemic control and cardiovascular risk factor control. The link worker model produced no

significant improvements in glycaemic control and only small improvements in cardiovascular risk factors. Further evaluation to determine the cost, effectiveness, and efficiency of these models in primary care settings through well-designed clinical trials is warranted. The adaptability of the 2 models from research settings and their potential application in other important chronic diseases should also be assessed. Either or both these models of care could then be developed to address the gap in service provision and optimize outcomes for minority ethnic groups.

Strengths and weaknesses of this review

We are unaware of any prior systematic reviews that have focused on processes within primary care that improve clinical outcomes in diabetes for minority ethnic groups. However, in a recent meta-regression analysis only 2 of 11 strategies for quality improvement—case management and team changes—showed modest improvements in glycaemic control among diabetic patients generally (Shojania et al., 2006). This systematic review should be interpreted within the context of several limitations. Although systematic reviews offer the least biased method of summarizing research evidence, all systematic reviews are limited by publication bias (Dickersin, 1990). Electronic searches of literature are also not exhaustive and some studies may not have been identified. The methodology and reporting of the included studies differed and was suboptimal for some. Furthermore, restricting to English language publications may have excluded some European studies. However, recent work has suggested that results of reviews with these limits do not differ substantially from reviews with no such limits (Burack et al., 1996).

We did not limit ourselves to examining only randomized controlled trials and, despite applying stringent quality criteria to studies, this may result in potential bias and confounding in the studies included. Conversely, since we included only randomized controlled trials and controlled clinical trials in this review, we excluded several studies with alternative designs that reported potentially worthwhile interventions using a broader approach to tackling the diabetes epidemic, for example, through media campaigns or religious groups (Yanek et al., 2001). Similarly, we were unable to include many studies in native or indigent groups. These could not be included because of the poor quality of the reporting of the findings or because of flaws in the design of the studies. Nevertheless, these novel approaches could make a significant impact at population level and need to be evaluated in future rigorous observational studies and clinical trials.

The heterogeneity of the study subjects, settings, design, and interventions limited what comparisons we could make from the studies and prevented us from carrying out a meta-analysis. Finally, few of the studies reported a comprehensive profile of impact of the intervention upon control of known risk factors for diabetic complications. Despite these limitations, this study provides a comprehensive review of the literature and our current understanding of the effects that attempt to improve diabetes care for minority ethnic groups.

Previous reports have focused on specific minority ethnic populations. For example, in the United Kingdom, most published literature on health status and health service use relates to South Asian and Afro-Caribbean populations (Gill et al., 2003) and in the United States much of the research is focused on African Americans with relatively fewer reports on Hispanic and other groups, in particular "native" or "indigenous" groups, for example, US and Canadian Native Indians known to have a high incidence of diabetes (Beach et al., 2004). The emergence of such groups as a minority is a function of historical and social processes albeit with common difficulties in access.

We recommend the following steps to improve the quality of research in this area. First, programs instituting changes should rigorously study these changes and report them to a rigorous standard, whether or not they demonstrate an impact (Des Jarlais et al., 2004). Second, interventions should be studied at multiple institutions to improve the

generalizability of the findings. Third, whenever possible, intervention studies should be conducted as randomized controlled trials to reduce the effect of confounding.

Clinical significance of this review

In systematic reviews of complex interventions, it is difficult to establish the active ingredients of an effective strategy and their individual contribution to clinical effectiveness and cost-effectiveness. Effectiveness may depend on how well interventions are implemented on the time frame over which implementation occurs. Nevertheless, this systematic review suggests that the case management model may be more suitable to the United States, while the link worker model is likely to be cheaper and more acceptable in the United Kingdom. To apply case management in other settings, we would need detailed information on the type of training and supervision given to case management workers. Moreover, the current paucity of nurses in some developed countries and in the United Kingdom may make recruitment of nurse case managers difficult (Buchan et al., 2003; UK tops league for international nurse recruitment, 2003). The only UK-based study that evaluated the case management approach found that it had no effect on patients' awareness of diabetes; thus, this approach would need a thorough evaluation before it can be implemented in the United Kingdom.

The role of link workers seems to be one of providing information, interpretation, and liaison with appropriate diabetes services, and mainly to improve diabetes awareness and self-management. The link worker approach has also demonstrated significant reductions in systolic and diastolic BP and a reduction in total cholesterol (O'Hare et al., 2004).

Many minority ethnic groups live in neighborhoods with high levels of deprivation, in large households, and on relatively low income (Beach et al., 2004; ONS, 2001). These are important factors that need to be accounted for when studying associations between ethnicity and ill health. Furthermore, the frequent lack of skills and resources within the health service to deal with this population is an impediment to both research and providing services for such groups. A detailed cost-effectiveness analysis would need to be incorporated into the further evaluation of both models, to gain information on both clinical effectiveness and cost-effectiveness for appropriate health policy decision making. In addition, to address concerns about whether any improvements in diabetic control are sustained, it would be useful to have a follow-up period of at least 24 months in future studies. However, both the case management and link worker models show promise and should be evaluated further as methods for improving diabetes outcomes in minority ethnic groups.

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